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Black Hole Horizons as Encoding Surfaces: Information, Thermodynamics, and the Resolution of the Information Paradox in the IAM Framework

Heath W. Mahaffey

Independent Researcher, Entiat, Washington, USA

hmahaffeyges@gmail.com github.com/hmahaffeyges/IAM-Validation

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Abstract

We present a thermodynamic analysis of black holes within the Informational Actualization Model (IAM), a gravitational decoherence framework validated against the Planck 2018 CMB likelihood across 17 converged MCMC chains. Three results are established. *First*, we demonstrate analytically that the Stefan-Boltzmann luminosity of the black hole horizon at the Hawking temperature reproduces the Hawking luminosity identically: $P_{\mathrm{SB}} = P_{\mathrm{Hawking}}$ exactly, for all masses. Hawking radiation is Stefan-Boltzmann radiation from a thermal encoding surface; the quantum field theory derivation and the thermodynamic derivation agree because they describe the same process. *Second*, we show that the black hole information paradox dissolves under examination of its premise: information is not stored in the interior volume but on the horizon surface $S_{\mathrm{BH}} = A/(4\ell_P^2)$, which has no independent interior degrees of freedom. Evaporation is the surface shrinking as information transfers from the hot local horizon (T_{BH}) to the cold cosmic horizon (T_{GH}) by ordinary thermodynamics. No information is destroyed; no firewall is required; no non-locality is needed. *Third*, we identify the universal horizon temperature formula $T = \hbar\kappa/(2\pi k_B)$ as the structural link connecting black hole physics to the IAM derivation of the electron rest mass presented in the companion paper [20]. Fundamental particles and black holes are fixed points of the same Landauer equation evaluated at their respective horizons, establishing IAM as a bridge between cosmology and particle physics. The Page curve is derived analytically from the IAM Landauer emission rate with no free parameters.

Keywords: black hole information paradox; Hawking radiation; Stefan-Boltzmann law; Landauer principle; holographic principle; Bekenstein entropy; Gibbons-Hawking temperature; Page curve; gravitational decoherence; M-sigma relation; JWST; information paradox resolution.

1. Introduction

The black hole information paradox has resisted resolution for fifty years despite contributions from Hawking, Susskind, Maldacena, Penrose, Polchinski, and many others [2, 3, 4, 5, 6, 17]. It is worth noting that Hawking himself substantially revised his position over the decades. His original 1975 argument held that thermal Hawking radiation destroys information, violating unitarity. By 2004 he conceded that information is likely preserved. In his final work — the "soft hair" papers with Perry and Strominger [1] — he argued that information is stored on the horizon surface itself, not in the interior, encoded in zero-energy quantum excitations he called soft hair. IAM does not contradict this final position. It provides the thermodynamic mechanism Hawking's work was pointing toward: a precise account of *why* information must be on the surface, grounded in the Landauer principle and gravitational decoherence.

The Informational Actualization Model [18] derives from the observation that gravitational decoherence — the irreversible conversion of quantum superpositions to classical outcomes — produces information at a thermodynamic cost governed by the Landauer principle [13]. This cost is paid at the cosmic Gibbons-Hawking temperature T_{GH} and encoded on the cosmic holographic boundary. The same mechanism, operating locally rather than globally, produces black holes when local decoherence exceeds the encoding capacity of the local bounding surface.

This paper establishes three results and draws one structural connection. Section 2 introduces the conceptual foundation: the distinction between entropy (pixel capacity) and information (pixel content), and the IAM picture in which matter does not enter a black hole but ceases to be projected at the horizon. Section 3 demonstrates $P_{\text{SB}} = P_{\text{Hawking}}$ analytically. Section 4 presents the IAM reframing of black holes as encoding surfaces. Section 5 dissolves the information paradox from its premise. Section 6 derives the Page curve. Section 7 identifies the universal temperature formula connecting black holes to the electron mass. Section 8 presents astrophysical applications.

2. Entropy, Information, and the Cessation of Projection

Before presenting the formal results, it is worth establishing a conceptual foundation that shapes how all subsequent results should be read. Two distinctions matter here, and conflating them has contributed to fifty years of confusion about what the black hole information paradox is actually asking.

2.1 Entropy is not information

In the context of black holes, these two words are often used interchangeably. They are not the same thing.

Entropy ($S_{\mathrm{BH}} = A/4\ell_P^2$) is a count of available encoding slots — the number of pixels on the holographic surface, the structural capacity of the boundary. It tells you how many distinct states *could* be encoded. It says nothing about what is actually written.

Information is what is actualized in those slots — the specific content of the final state, the record of which quantum superpositions were resolved and how. In IAM language: entropy is the pixel array; information is what is written on the pixels.

Decoherence is the act of writing: it converts quantum potential into a classical actualized state at the Landauer cost of $k_B T \ln 2$ per bit. The Landauer principle bridges entropy and information exactly — each bit written costs one entropy unit of energy at the local temperature.

This distinction matters because it changes what we think is stored on the black hole horizon. The Bekenstein-Hawking entropy $S_{\mathrm{BH}} = A/4\ell_P^2$ is the *capacity* of the horizon — the number of pixels available. The information encoded in those pixels is the specific actualized state of the last decoherence event before the projection was severed. The horizon does not store information in the way a hard drive stores a file. It stores the entropy structure — the pixel pattern — that *is* that information, frozen at the moment projection ceased.

2.2 Matter does not enter a black hole — the projection ceases

This is the central physical claim of this paper, and it dissolves the information paradox more completely than any mechanism that accepts the premise of interior storage.

In IAM, a particle exists as a stable, self-consistent decoherence loop. Three elements are required to maintain it:

1. *Pure potential* — the quantum superposition, the unactualized field state.
2. *The projector* — gravitational decoherence, the process that converts potential to actualized classical outcome at the Landauer cost.
3. *The screen* — the cosmic holographic boundary, the encoding surface on which the actualized state is registered. Without registration, the decoherence loop cannot be maintained.

A particle exists because its decoherence loop is continuously refreshed and registered on the cosmic encoding surface. The Landauer cost is the energy required to keep the projector running. The rest mass of the particle *is* this cost, evaluated at the Gibbons-Hawking temperature — which is precisely what the electron mass fixed-point equation in the companion paper [20] expresses.

When a particle approaches a black hole horizon, something specific happens: the cosmic encoding surface becomes causally inaccessible. The interior of a black hole is not connected to the cosmic horizon by any causal path. The screen goes dark.

The projector cannot reach the screen. The decoherence loop cannot be registered. The particle *ceases to be projected.*

It does not travel inward and carry its information with it. It does not get copied onto the horizon by some exotic mechanism. The loop that constituted it simply cannot be sustained in a region where the cosmic encoding surface is inaccessible.

What appears on the black hole horizon is not the particle. It is the *entropy imprint* of the last actualized state — the final frame of the projection before the screen went dark. This entropy imprint, frozen in the pixel structure of the horizon at $S_{\mathrm{BH}} = A/4\ell_P^2$, encodes the informational content of that final state. The Bekenstein-Hawking entropy is not a measure of hidden interior degrees of freedom. It is the area of the last frame.

This picture is consistent with — and provides the mechanism for — Hawking's final "soft hair" result [1]. Soft hair is the pixel structure of the last frame, written in zero-energy quantum excitations on the horizon surface. IAM explains why that structure must be there: the Landauer principle requires that every decoherence event be registered on an encoding surface, and the horizon is the last surface the projector could reach.

The information paradox, in this picture, is not a paradox about escaping information. It is a question about what it means for matter to "enter" a black hole at all. The answer IAM offers is: it does not. The projection stops. The entropy imprint remains. That imprint is slowly radiated away as

Hawking radiation as the horizon shrinks — not because information escapes from the interior, but because the last frame is being erased one pixel at a time, each erasure costing $k_B T_{\mathrm{BH}} \ln 2$ of energy, exactly as Landauer requires.

3. Hawking Radiation Is Stefan-Boltzmann Radiation

The Hawking temperature of a Schwarzschild black hole of mass M is:

$$T_{\mathrm{BH}} = \frac{\hbar c^3}{8\pi G M k_B} \tag{1}$$

The Stefan-Boltzmann luminosity of a blackbody surface at temperature T with area A is $P = \sigma_{\mathrm{SB}} A T^4$, where $\sigma_{\mathrm{SB}} = \pi^2 k_B^4/(60\hbar^3 c^2)$. The black hole horizon area is $A_{\mathrm{BH}} = 16\pi G^2 M^2/c^4$.

Result 1 (Stefan-Boltzmann = Hawking)**.** *Substituting T_{BH} and A_{BH} into $P_{SB} = \sigma_{SB} A T^4$:*

$$P_{SB} = \frac{\pi^2 k_B^4}{60\hbar^3 c^2} \cdot \frac{16\pi G^2 M^2}{c^4} \cdot \left(\frac{\hbar c^3}{8\pi G M k_B}\right)^4 = \frac{\hbar c^6}{15360\pi G^2 M^2} = P_{Hawking} \tag{2}$$

The ratio $P_{SB}/P_{Hawking} = 1.000000$ for all M. $\square$

This identity is not a coincidence or consistency check. It means that Hawking radiation *is* thermal blackbody radiation from a hot encoding surface. The quantum field theory derivation (Hawking 1975) is the microscopic explanation; the Stefan-Boltzmann law is the macroscopic limit. They agree because they describe the same physical process: a thermal surface at temperature T_{BH} radiating energy at the Landauer cost of $k_B T_{\mathrm{BH}} \ln 2$ per bit.

The Landauer encoding rate of the horizon follows immediately:

$$\Gamma_{\mathrm{BH}} = \frac{P_{\mathrm{Hawking}}}{k_B T_{\mathrm{BH}} \ln 2} = \frac{c^3}{1920\, G M \ln 2} \quad \text{[bits per second]} \tag{3}$$

For reference: a 1 $M_\odot$ black hole encodes at $\Gamma \approx 152$ bits/s; a $10^9\, M_\odot$ SMBH at $\Gamma \approx 1.5 \times 10^{-7}$ bits/s. Smaller, hotter black holes encode faster.

4. Black Holes as Encoding Surfaces

4.1 The Reframing

Standard picture: gravity collapses matter; extreme curvature creates a one-way membrane; nothing escapes.

IAM picture: gravitational decoherence accelerates during collapse; information production rate exceeds local encoding capacity; the universe resolves the encoding deficit by creating a new surface. The horizon is not a barrier — it is an *encoding surface* created by thermodynamic necessity.

This reframing has a specific consequence: the black hole horizon is not the boundary between "inside" and "outside." It is the storage medium. The information is not *behind* the horizon — it is *on* the horizon.

4.2 The Saturation Condition

The Bekenstein-Hawking entropy $S_{\mathrm{BH}} = A/(4\ell_P^2)$ is a *surface* quantity, not a volume quantity. The maximum information storable in any region is bounded by the area of its boundary (holographic principle [9, 10]), not its volume. The interior of a black hole has no independent degrees of freedom beyond those encoded on the horizon.

This is the resolution to the information paradox stated as a *premise correction* rather than a mechanism: information does not enter the interior and require extraction. Information is written onto the horizon surface as matter crosses it, at the Landauer cost of $k_B T_{\mathrm{BH}} \ln 2$ per bit. There is no interior information to recover.

4.3 Two Regimes of the Same Mechanism

IAM's gravitational decoherence feedback loop operates in two regimes:

	Cosmological (self-regulating)	Collapse (runaway)
Decoherence rate	Gradual, from structure formation	Explosive, from gravitational collapse
Information encoding	Distributed on cosmic horizon	Concentrated on local horizon
Pressure release	Cosmic expansion dilutes matter	No valve — gravity wins locally
Result	$\mu < 1$, $\Sigma = 1$, $E(a) \to e$	New encoding surface: black hole

Same physics. Same mechanism. Different boundary conditions.

5. Resolution of the Information Paradox

5.1 The Paradox and Its Premise

Hawking [2] showed that black holes emit thermal radiation. Thermal radiation is random: it carries maximum entropy and no information about what produced it. If a black hole fully evaporates via thermal Hawking radiation, the information about its formation is gone. This violates quantum mechanical unitarity.

Every proposed resolution — complementarity [11], AdS/CFT [4], firewalls [5], ER=EPR [6], island formula [7, 8] — accepts this premise and works out machinery to extract the information from the interior.

5.2 The Premise Is Incorrect

Proposition 1 (No interior degrees of freedom)**.** *The interior of a black hole has no degrees of freedom independent of those encoded on the horizon surface.*

Argument. The Bekenstein-Hawking entropy $S_{\mathrm{BH}} = A/(4\ell_P^2)$ is a *surface integral*. It scales with area, not volume. The holographic principle [9] establishes that the maximum information content of any region is bounded by its boundary area. For a black hole, the boundary is the horizon. The interior is the holographic projection of the boundary state, not an independent information storage system. There is no interior information to be lost. □

5.3 The Dissolution

If information is on the surface, then evaporation is the surface shrinking. As the black hole radiates:

- M decreases; A_{BH} decreases; S_{BH} decreases.
- Each bit of surface entropy is paid off in the Hawking luminosity at the Landauer cost $k_B T_{\mathrm{BH}} \ln 2$.
- The information is not "escaping" the interior — the surface is transferring it to the outgoing radiation.

The radiation is not truly thermal in the information-theoretic sense. It is thermal *at each instant*, but the temperature $T_{\mathrm{BH}}(M)$ rises as M decreases, so the time series of Hawking emissions carries a systematic frequency shift that encodes the evaporation history. The information is in the correlations across time, not in individual photon energies.

No information is destroyed. No unitarity is violated. No firewall is required. No non-locality is needed. The paradox dissolves because its premise — that information is interior — is not correct.

5.4 Why the Firewall Does Not Arise

The AMPS argument [5] shows that if early Hawking radiation carries information, quantum monogamy is violated: the outgoing radiation cannot be entangled with both the early radiation (required for unitarity) and the infalling modes (required by the equivalence principle).

The IAM resolution eliminates one of the three parties. The infalling modes are not a separate system — they are encoded on the horizon surface at the moment of crossing. There is no independent interior mode to participate in the monogamy paradox. The AMPS trilemma has only two parties, which is not a paradox.

6. The Page Curve from IAM

The Page curve [3] describes how the von Neumann entropy of Hawking radiation evolves: it should rise to $t_{\mathrm{Page}} = t_{\mathrm{evap}}/2$ then fall back to zero for a unitary evaporation. This "unitarity test" distinguishes information-carrying from truly random radiation.

In IAM, the entropy of Hawking radiation tracks the horizon entropy being transferred to the radiation:

$$\frac{dS_{\mathrm{rad}}}{dt} \;=\; \Gamma_{\mathrm{BH}}(M(t)) \;=\; \frac{c^3}{1920\, GM(t) \ln 2} \tag{4}$$

The mass evolves as:

$$\frac{dM}{dt} \;=\; -\frac{P_{\mathrm{Hawking}}(M)}{c^2} \;=\; -\frac{\hbar c^4}{15360\pi G^2 M^2} \tag{5}$$

Equation (5) integrates analytically to:

$$M(t)^3 \;=\; M_0^3 - \frac{\hbar c^4 t}{5120\pi G^2} \qquad \Longrightarrow \qquad t_{\mathrm{evap}} = \frac{5120\pi G^2 M_0^3}{\hbar c^4} \tag{6}$$

Result 2 (IAM Page Curve)**.** *Substituting equation (6) into (4) and integrating:*

$$S_{rad}(t) = \int_0^t \frac{c^3}{1920\, GM(t') \ln 2}\, dt' \tag{7}$$

The radiation entropy rises as M decreases (hotter BH encodes faster), reaching $S_{BH,0}/2$ at $t = t_{evap}/2$ (the Page time), then completing the transfer as $M \to 0$. The total radiation entropy at t_{evap} equals the initial horizon entropy $S_{BH,0}$: all bits transferred, none created, none destroyed. Unitarity is preserved.

For a solar-mass black hole: $t_{\mathrm{evap}} \approx 2.1 \times 10^{58}$ Gyr, $S_{\mathrm{BH},0} \approx 2.6 \times 10^{76}$ bits, $\Gamma_{\mathrm{BH}} \approx 152$ bits/s. The curve shape is universal; the timescale scales as M_0^3.

7. The Universal Temperature Formula: Bridge to Particle Physics

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Equations (1) and (**??**) share a single structure:

$$T \;=\; \frac{\hbar\kappa}{2\pi k_B} \tag{8}$$

where κ is the surface gravity of the relevant horizon:

Horizon	Surface gravity κ	Temperature
Black hole	$c^3/(4GM)$	$T_{\mathrm{BH}} = \hbar c^3/(8\pi GMk_B)$
Cosmic (de Sitter)	H_0	$T_{\mathrm{GH}} = \hbar H_0/(2\pi k_B)$
Unruh (Rindler)	a/c	$T_U = \hbar a/(2\pi ck_B)$

The 2π in each case arises from the same source: the angular periodicity of the 2D holographic boundary encoding. This is not a coincidence of units — it is a statement that all horizons are thermal encoding surfaces obeying the same geometry.

The fixed-point connection. The companion paper [20] shows that the electron rest mass satisfies a Landauer fixed-point condition at the *cosmic* horizon:

$$m_e c^2 \;=\; k_B T_{\text{GH}} \ln 2 \;\times\; \frac{(m_P/m_e)^{3/2}}{\alpha^{5/2}} \tag{9}$$

An identical condition at the *local* horizon gives:

$$M c^2 \;=\; k_B T_{\text{BH}} \ln 2 \;\times\; S_{\text{collapse}}^{1/2} \tag{10}$$

Equations (9) and (10) are the same fixed-point condition $[\text{rest energy} = \text{Landauer cost} \times N]$ evaluated at different horizons. The electron is the cosmic horizon's fixed-point particle. The black hole is a local horizon's fixed-point object. Fundamental particles and black holes belong to the same class.

This structural unity means that:

- The encoding throughput formula $\Gamma_{\text{BH}} = c^3/(1920GM \ln 2)$ is the template for deriving the $\alpha^{5/2}$ suppression in the electron mass equation (Open Problem 1 of [20]).
- The 2π in T_{BH} is the same 2π in T_{GH}, which is the same 2π in $(2\pi)^{-1/10}$ (Open Problem 2 of [20]).
- Completing the formal derivations in the particle physics paper simultaneously deepens the black hole thermodynamic framework.

The two papers are not independent. They are two descriptions of one mechanism.

8. Astrophysical Applications

8.1 The M-σ Relation as a Fixed Point

The observed $M_{\text{BH}} \propto \sigma^4$ relation between SMBH mass and galaxy velocity dispersion is established observationally but lacks a clean theoretical derivation.

In IAM, the SMBH is the fixed-point encoding surface for its host galaxy. The galaxy's gravitational decoherence rate scales as σ^4 (from the virial theorem applied to the stellar velocity dispersion). The SMBH encodes this at the Landauer rate $\Gamma_{\mathrm{BH}}(M)$. Equilibrium requires these rates to match, giving $M \propto \sigma^4$ — the M-σ relation as a thermodynamic fixed point of the same equation that determines the electron mass.

8.2 JWST Early Black Holes

JWST has identified massive black holes at $z > 7$, earlier than standard accretion models easily explain. In IAM, black holes are the universe's primary encoding infrastructure before large-scale structure formation makes the cosmic horizon dominant. Early, massive BH formation is not a puzzle — it is a prediction. The universe front-loads its encoding capacity because BHs are the most efficient encoding surfaces available in the early universe.

The IAM prediction: early BH masses should correlate with the local information production environment (density, interaction rate) rather than accretion history alone. The M-σ relation should evolve with redshift in a specific way tied to the transition from BH-dominated to cosmic-horizon-dominated encoding.

8.3 Cosmic Censorship as Landauer Necessity

Penrose's cosmic censorship conjecture — that naked singularities cannot form — is an independent postulate in GR. In IAM it is a consequence of the Landauer principle: a naked singularity would be a region of infinite information production rate with no encoding surface. Landauer's principle is not optional: information must be encoded. The universe must provide a horizon. Cosmic censorship is thermodynamically enforced, not geometrically postulated.

9. Comparison with Existing Resolutions

Resolution	Core mechanism	IAM comparison
Complementarity [11]	Observer-dependent description	IAM: simpler — info on surface for all observers
AdS/CFT [4]	Boundary encodes bulk (AdS)	IAM: same conclusion in de Sitter (our universe)
Firewall [5]	Drama at horizon preserves unitarity	IAM: no interior d.o.f., firewall unnecessary
ER=EPR [6]	Hawking pairs via wormholes	IAM: no wormholes, ordinary thermodynamics
Island formula [8]	Replica topology, Page curve	IAM: same Page curve from Landauer rate, no topology

The IAM resolution is the most parsimonious: it requires no new mathematical structures, no exotic topology, no modifications to quantum mechanics, and no non-locality. It requires only that the holographic principle be taken seriously — that the horizon surface is the storage medium, not a boundary concealing interior storage.

10. Discussion and Conclusion

We have established three results within the IAM framework: (1) $P_{\text{SB}} = P_{\text{Hawking}}$ exactly, confirming the BH horizon as a thermal encoding surface; (2) the information paradox dissolves when the premise that information is interior is replaced by the correct holographic statement that information is on the surface; (3) the Page curve follows analytically from the Landauer emission rate with no free parameters.

The structural result is the universal temperature formula $T = \hbar\kappa/2\pi k_B$, which connects black hole physics to the electron mass derivation of the companion paper [20]. Black holes and fundamental particles are fixed points of the same Landauer equation at different horizons. This is not metaphor — it is the same equation, the same 2π, the same Bekenstein saturation condition, evaluated at local vs. cosmic scales.

IAM is a bridge. The cosmological community has MCMC-validated predictions for Euclid and DESI. The particle physics community has an electron mass derivation and a Koide formula mechanism. The black hole community has a thermodynamic dissolution of the information paradox. These are not three separate results — they are one mechanism seen from three angles.

The most testable near-term prediction: photonic qubits, living in the $\Sigma = 1$ sector, pay no Landauer cost for maintaining superposition. Matter qubits do. Decoherence rates in matter qubits should show a component scaling with local gravitational potential; photonic qubit decoherence rates should not. This is a specific, measurable prediction distinguishing IAM from standard quantum mechanics.

All MCMC data, CAMB modifications, and analysis scripts are publicly available at

`github.com/hmahaffeyges/IAM-Validation`.

10. References

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[1] Hawking, S. W., Perry, M. J., & Strominger, A. (2016). Phys. Rev. Lett., **116**, 231301.

[2] Hawking, S. W. (1975). Commun. Math. Phys., **43**, 199.

[3] Page, D. N. (1993). Phys. Rev. Lett., **71**, 3743.

[4] Maldacena, J. (1998). Adv. Theor. Math. Phys., **2**, 231.

[5] Almheiri, A., Marolf, D., Polchinski, J., & Sully, J. (2013). JHEP, **2013**(02), 062.

[6] Maldacena, J. & Susskind, L. (2013). Fortschr. Phys., **61**, 781.

[7] Penington, G. (2020). JHEP, **2020**(09), 002.

[8] Almheiri, A. et al. (2020). JHEP, **2020**(05), 013.

[9] 't Hooft, G. (1993). gr-qc/9310026.

[10] Susskind, L. (1995). J. Math. Phys., **36**, 6377.

[11] Susskind, L., Thorlacius, L., & Uglum, J. (1993). Phys. Rev. D, **48**, 3743.

[12] Bekenstein, J. D. (1973). Phys. Rev. D, **7**, 2333.

[13] Landauer, R. (1961). IBM J. Res. Dev., **5**, 183.

[14] Gibbons, G. W. & Hawking, S. W. (1977). Phys. Rev. D, **15**, 2738.

[15] Jacobson, T. (1995). Phys. Rev. Lett., **75**, 1260.

[16] Cai, R.-G. & Kim, S. P. (2005). JHEP, **2005**(02), 050.

[17] Penrose, R. (2004). The Road to Reality. Jonathan Cape.

[18] Mahaffey, H. W. (2026). IAM theoretical derivation. GitHub: hmahaffeyges/IAM-Validation.

[19] Mahaffey, H. W. (2026). Submitted: Universe, ID: universe-4189350.

[20] Mahaffey, H. W. (2026). Electron mass and Koide formula from holographic horizon thermodynamics. GitHub preprint.

[21] Mahaffey, H. W. (2026). Virial partition across 37 orders. GitHub: hmahaffeyges/IAM-Validation.

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